

current wasted is minimised, raising the conversion efficiency from 31.5 per cent to 35.8 per cent.

Thin-film substrates

Thin-film substrates are gaining popularity, as an alternative to silicon wafers, in the production of solar cells. These substrates overcome the cost constraint of silicon wafers. Moreover, the thin-film technology also enables solar cells to be constructed on flexible substrates, making it possible to incorporate solar cells into varied objects.

Around November last year, DEK Solar announced that its PV1200 photovoltaic metallisation platform could be used not just for silicon wafers but also to meet the commercial challenges of thin-film substrates. In association with key industry players, DEK has developed a range of metallisation solutions for thin-film substrates—including precision deposition of silver epoxy compounds for metallisation, application of other active and interconnect layers, printing techniques similar to those already proven in bulk silicon solar cell production, etc.

The PV1200 delivers 1200 cells-per-hour throughput, six-sigma process rating, 12.5 micron resolution and advanced handling capabilities.

Solar ink for better efficiency

At the Solar Power International 2008 held in October, Five Star Technologies unveiled its ElectroSpense S-series solar inks for use in crystalline silicon photovoltaic cells. This ink helps solar cell manufacturers by maintaining consistent electrical contact with emitter layers, thereby improving manufacturing yields and boosting overall cell efficiency.

These inks are produced using a hydrodynamic cavitation process that enables high silver and glass particle uniformity and particle sizes in the low to sub-micrometre range. This allows solar cell makers to layout front-side grid lines as narrow as 80µ to 100µ, which is much thinner than the 120µ to

Solar Energy in India

There are 86,000 villages in India. Each household spends about US\$ 5-8 a month on kerosene for lighting their homes. Whereas, by installing a 70-watt solar panel, an entire household can generate electricity in less than an instalment of US\$ 5 every month. This 70-watt solar panel on being charged for five hours can provide 350 watt-hour of energy a day, and can run an entire Indian household.

For the solar dream to become reality, it is essential that different ecosystems rally together to innovatively reduce the cost per watt to make it a viable option for all.

To realise the importance and relevance of solar power in India we must consider the following parameters:

Scalability. 5000 trillion kW-h energy incidence over India with most parts receiving 4-7 kW-h per sq metre per day. Hence, solar thermal and solar photovoltaic can be harnessed. Photovoltaic can especially work effectively for rural electrification as well as diffused radiation conditions.

Environment impact. Solar energy scores high on environmental impact with close to zero emissions. This could also lead to a reduction of 42 million tonne of carbon dioxide emission, thereby saving on precious fuel oil. Also, the on-grid connectivity enables feeding of power into the grid to access and push power into the grid based on net metering (Feed-in-Tariff, or FIT). On the other hand, off-grid applications do not require the infrastructure related to transmission.

Cost. Solar energy is currently high on absolute costs compared to coal. The objective of the mission is to drive down the costs to Rs 5-6 per kWh by 2017-20 and make it competitive with fossil fuels.

"Government and corporate India must spread awareness about solar products. With a population of more than 1 billion, India has a huge untapped market that can choose to use energy such as solar," remarks Dr Atre.

—Dr Madhusudan Atre, President, Applied Materials India

Solar Energy Companies in India

Source Guides List of Manufacturers

'<http://energy.sourceguides.com/businesses/byGeo/byC/India/byP/solar/pvC/byB/mfg/mfg.shtml>

National Informatics Centre (NIC) List of Photovoltaic Manufacturers in India

'<http://mnes.nic.in/list/spv-manufacturers.pdf>

Indian Solar Industry Companies

'<http://www.solarbuzz.com/CompanyListings/India.htm>

175µ width of conventional contacts.

Alternatives of silicon

In July this year, a team of scientists at the University of California, Los Angeles, announced that their breakthrough could significantly reduce the price of solar technologies, by using materials less expensive than silicon. This could potentially end a long-drawn search for materials that are more efficient, cheaper to produce and use fewer raw materials than silicon. Tests of copper, indium, gallium, selenide, etc, have failed so far.

The team at UCLA, however, has demonstrated a low-cost processing method for solar cells made from copper, indium and diselenide. Those cells, they claim, will have the potential to be

produced on a large scale for a number of applications, including placement on backpacks or clothing.

So far, the researchers have achieved 9.13 per cent efficiency over the 16-month project, but they are confident, as said to a New York Times reporter, that they will reach their goal of 15 or 20 per cent efficiency.

According to the report, "Most CIGS solar cells are produced using a co-evaporation technique that involves vacuum and can be costly and time-consuming. The elements are heated and deposited on a surface in a vacuum. But the UCLA team has created its copper-indium-diselenide solar cell without going through the vacuum evaporation process. Instead,